Huawei Access Node KR-ACC-15 QoS Congestion Event Resolution SOP

Node: KR-ACC-15 | **Vendor:** Huawei | **Severity:** Minor | **Alarm Code:** ALARM_5123

Description

This Standard Operating Procedure (SOP) addresses a QoS congestion event on Huawei Access Node KR-ACC-15, Site Location KR Puram Access, and Role Access. The failure domain in question is a minor severity ALARM_5123, which is indicative of a congestion event involving microbursts, interface tail drops, and QoS queue starvation. This phenomenon arises from the inherent nature of the Huawei Access Node's hardware and software architecture, particularly the implementation of the IEEE 802.1Q-2014 standard for VLAN tagging and the associated Quality of Service (QoS) mechanisms. The Huawei Access Node's hardware is designed to handle a high volume of traffic, but in the event of a sudden spike in traffic, the node's buffers can become overwhelmed, leading to a congestion event. Furthermore, the node's QoS mechanism, which is based on the DiffServ model, can also contribute to congestion by prioritizing traffic based on its DSCP (Differentiated Services Code Point) value, potentially causing high-priority traffic to be dropped when the buffers are full, leading to a queue starvation condition.

Pre-requisite Checks

- Ensure that the Huawei Access Node KR-ACC-15 is reachable via the network management system (NMS) and that the node is properly powered on.

- Verify that the redundant link is up and operational.

- Dispatch remote hands to the KR Puram Access site to visually inspect the node and confirm that there are no physical issues with the node or its connections.

Potential Root Causes

Cause	Probability
Buffer Overflow Due to Sudden Traffic Spike	High
QoS Queue Starvation Due to High-Priority Traffic Dropping	Medium
Interface Tail Drops Due to Congestion	Low

Diagnostic Commands & Expected Outputs

```
Command: display interface all, Expected Output: Interface statistics, including errors and drops., Description: This command displays the statistics for all interfaces on the Huawei Access Node, including errors and drops, which can help identify if there are any issues with the interfaces that may be contributing to the congestion event.
```

```
Command: display qos queue, Expected Output: QoS queue statistics, including queue size and drop count., Description: This command displays the QoS queue statistics, including queue size and drop count, which can help identify if there are any issues with the QoS mechanism that may be contributing to the congestion event.

Command: display buffer, Expected Output: Buffer statistics, including buffer size and drop count., Description: This command displays the buffer statistics, including buffer size and drop count, which can help identify if there are any issues with the buffers that may be contributing to the congestion event.
```

Resolution Steps

Step 1: Step: Clear Buffer Overflow, Command: clear buffer, Description: This command clears the buffer overflow condition on the Huawei Access Node, which can help resolve the congestion event.

Step 2: Step: Adjust QoS Queue Settings, Command: qos queue {queue-id} {new-queue-size}, Description: This command adjusts the QoS queue settings to increase the queue size, which can help prevent queue starvation and congestion events.

Step 3: Step: Verify Interface Configuration, Command: display interface {interface-id}, Description: This command verifies the interface configuration to ensure that it is properly configured and not contributing to the congestion event.

Post-Check Verification

```
Command: display interface all, Expected Output: Interface statistics, including errors and drops., Description: This command verifies that the interface statistics are within normal range and that there are no errors or drops.

Command: display qos queue, Expected Output: QoS queue statistics, including queue size and drop count., Description: This command verifies that the QoS queue statistics are within normal range and that there are no drops.

Command: display buffer, Expected Output: Buffer statistics, including buffer size and drop count., Description: This command verifies that the buffer statistics are within normal range and that there are no drops.
```

Roll-back Procedures

- Step: Restore Original QoS Queue Settings, Command: undo qos queue {queue-id} {new-queue-size}, Description: This command restores the original QoS queue settings in case the new settings do not resolve the congestion event.

- Step: Restore Original Buffer Settings, Command: undo clear buffer, Description: This command restores the original buffer settings in case the clear buffer command does not resolve the congestion event.